

Human Development, Organ Donation and Transplant Efficiency

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1. Research Aim

The purpose of this report is to investigate whether the degree of human development in a country is related to organ donation capacity, transplant activity, and utilisation efficiency. One of the analyses focuses on the Human Development Index (HDI), a composite indicator developed by the United Nations to provide a summary of average achievement in three major areas of human development: health, education, and standard of living. More specifically, HDI is a composite of life expectancy, years of schooling, and gross national income per capita that can be utilised to compare countries on a broader basis.^[1] To this end, the report attempts to determine whether countries with higher HDI values have better donation systems, whether their transplant systems are also structurally different, and whether HDI can adequately explain utilisation efficiency. This study uses international data from the WHO Global Observatory on Donation and Transplantation^[9] and HDI data from UNDP/Our World in Data.^[1] After the harmonisation process, the analysed sample includes 122 countries for donation and transplant analysis and 88 countries with active deceased-donor programmes for efficiency analysis.

2. Data Preparation and Analytical Strategy

2.1 Construction of the tidy dataset

Before statistical analysis, the original WHO spreadsheet was reorganized based on tidy-data guidelines to ensure that a dataset can be analyzed consistently.^[4] Because donor activity and transplant activity represent separate observational units, two separate tidy tables were created: `donors_tidy` and `transplants_tidy`. The spreadsheet was then transformed with a melting operation from wide to long format, and names of compound columns were separated into meaningful analytical variables. Aggregate total columns were not included in the final tidy tables when they were simple sums of more detailed categories, but they were kept in the raw data for validation. This step was necessary to move the data storage from a spreadsheet to a format suitable for statistical analysis.^[4]

2.2 Variables and statistical methods

Four primary analytical variables were developed: deceased donors per million population (PMP), transplants per million population, utilisation efficiency, and HDI tier. Linear association was evaluated with Pearson's r ^[10] and Spearman's ρ ^[5] to capture monotonic relationships and to minimise susceptibility to outliers. LOESS regression was also calculated to detect the countries that were performing above or below the level predicted by HDI through the residuals.^[6] In the missingness plot against HDI, the blue line is the fitted LOESS regression line, included to summarise the average linear association between HDI and the share of missing data. All p-values are two-tailed.

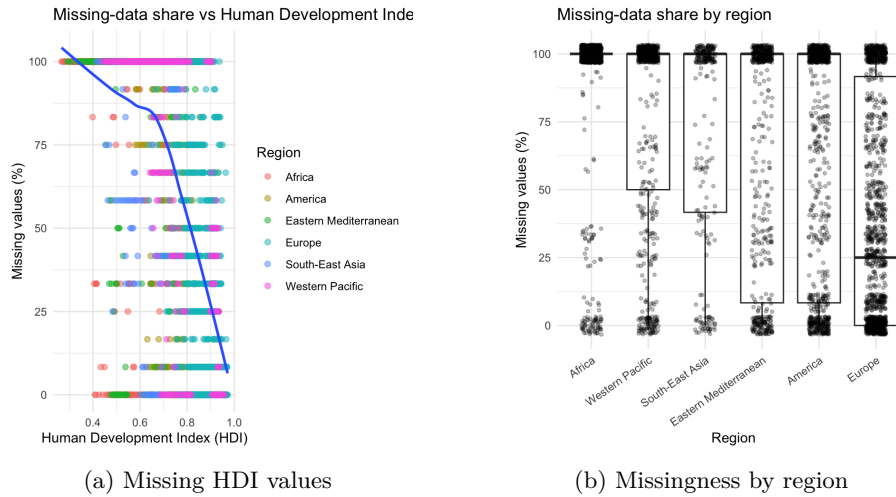


Figure 1: Coverage diagnostics for the merged WHO-HDI dataset.

3. Main Findings

3.1 Missingness, harmonisation, and sample coverage

High missingness is one of the main limitations of the dataset. The missing values were not treated mechanically. Instead, a distinction was made between observed zero values, unreported values, and structurally unavailable values. Reported zeros were retained as true zero activity, while NA values remained missing or unreported observations that were not converted into zero. This choice is important because replacing all data gaps with zero would confound the absence of recorded activity with the absence of reporting and would therefore bias country comparisons.

According to Rubin's framework, missingness can be classified as MCAR, MAR, or MNAR.^[3] As shown in Figure 1, the visual distribution of missing HDI values and the concentration of missing observations in some regions suggest that missingness

may be structured rather than uniformly random. This is consistent with the more recent discussion by Lee et al., who argue that missing-data handling should be planned in relation to the data-generation process rather than treated as a purely technical cleaning step.^[2] For this reason, the analysis preserves missingness as information and standardises country names before merging, since the WHO and HDI datasets adopted different naming conventions for 21 countries.

Figure 1 supports the same inference: missingness is structured rather than uniformly distributed. The missing-HDI image highlights the reduction in the usable sample after harmonisation, while the regional image suggests that some areas contribute more incomplete observations than others. Methodologically, this justifies avoiding automatic imputation and interpreting the final sample as the result of both substantive differences in reporting capacity and technical differences in data availability.^[2,3]

3.2 HDI and donation/transplant capacity

The most explicit result of the analysis is that HDI is strongly linked to organ donation and transplant performance. Countries with higher HDI values tend to have more deceased donors per million population and more transplants per million population, a pattern later summarised in panel (a) of Figure 2. There is a statistically significant relationship between them in each of these cases. For donor rates, the Pearson correlation with HDI is $r = 0.62$, using Spearman $\rho = 0.75$. The transplant rate is $r = 0.68$, $\rho = 0.77$ (all p-values less than 0.001). It is also interesting to note the differences between Pearson and Spearman coefficients. Since Spearman’s ρ is higher than Pearson’s r , the relationship seems to be stronger in rank terms than strictly linear terms. In practical terms, this indicates that transplant capacity does not increase consistently with development. Instead, it rises relatively increasingly once countries reach greater development levels, especially past 0.80 HDI value. This interpretation agrees with the comparison among HDI tiers, in which the Very High group has much higher median donor and transplant rates than the other groups. Overall, human development appears to provide the structural conditions under which donation systems can expand in a substantial way.

4. Results Overview and Interpretation

Figure 2 summarises the four main empirical findings discussed in the rest of this section.



Figure 2: Relationship between HDI and organ donation outcomes.

The pooled results image is a four-part summary of the empirical results. As shown in panel (a), increased HDI levels correlate with increased donor and transplant rates. Panel (b) illustrates how low-HDI countries depend substantially more on living donation, whereas deceased donation becomes dominant in higher-HDI systems. Panel (c) indicates that the mix

of transplanted organs becomes more diverse as development increases, which suggests greater system maturity. Finally, panel (d) shows that some countries perform better than their HDI alone would indicate. Throughout this summary, HDI is interpreted consistently with the Human Development Index reference as a composite measure of health, education, and standard of living.

The change illustrated in panels (b) and (c) is not only quantitative, but institutional. A system based largely on deceased donation necessitates intensive care capacity, donor-identification processes, transplant coordination, legal recognition of brain death, and organisational facilities providing organs for fast and safe transportation. The image thus backs up the conclusion that HDI is connected to transplant outcomes in part because it is a measure of wider health-system capacity.

The panel showing an outlier is particularly informative for interpretation because it demonstrates that development is only one of the factors that affect the outcome. Spain overperforms against its HDI when it combines favourable structural conditions with a highly coordinated donation system, specialist transplant coordinators, and a well-established organisational model that actively identifies potential donors in hospitals rather than relying on passive willingness to donate. These institutional features help explain why Spain performs above the level predicted by HDI alone.

Meanwhile, Japan underperforms relative to its HDI despite its very high level of development, suggesting that the limiting factors are institutional and cultural rather than economic. The organ donation literature notes that low deceased-donor rates may persist where brain-death criteria remain contested, family-consent practices are restrictive, and public attitudes toward post-mortem donation are cautious.^[7,11] In this sense, Japan supports the broader conclusion of the paper: high development creates capacity, but law, coordination, trust, and cultural acceptance determine whether that capacity is translated into deceased donation.

The residual structure is also informative. Overperformers are not evenly distributed across the sample: many are European countries with established deceased-donor programmes, whereas some of the clearest underperformers are wealthy Asian and Gulf states. This geographical clustering suggests that policy diffusion, legal frameworks, and organisational culture matter in ways that HDI alone does not capture. Development also seems to create the capacity for donation but national systems still require rules, routines and public legitimacy to convert that capacity into actual donors.

In panel (d), Spain lies roughly 25 donors PMP above the value predicted by the HDI-based model, whereas Japan lies more than 10 donors PMP below it, creating a gap of nearly 40 donors PMP between two very high-development settings. This distance suggests that policy design can be as substantively important as development itself once countries enter the upper HDI range. The strong performance of countries such as Croatia and Mali also indicates that countries do not need to be the richest in the sample to exceed expectations: coherent coordination, professionalised donor detection, and supportive social norms may help compensate for broader structural constraints over time.

Cultural, political, and institutional conditions appear to contribute substantially to the results in panel (d). The figure does not establish which mechanism is dominant, but it is consistent with variation in consent regimes, public trust in health institutions, acceptance of brain death, and the extent to which hospitals are organised to identify prospective donors. Methodologically, this part of the analysis should still be interpreted as suggestive rather than causal, as these bars are residuals from a simple linear model rather than estimates from a full causal design.

Spain's leading position is best explained not simply by development, but by its ability to translate capacity into donation through institutional design. The comparative literature often attributes this to the Spanish model, which combines strong national coordination, transplant coordinators at multiple levels of the health system, active donor identification in hospitals, and established clinical routines for approaching potential donation opportunities. In the context of panel (d), this helps explain why Spain sits above the level predicted by development alone and why institutional organisation appears central to differences between otherwise highly developed countries.^[8]

A further outcome that is drawn from the image is that utilisation efficiency follows a different logic compared to donation volume. In the latest-year-per-country dataset, Pearson's correlation between HDI and efficiency is almost zero ($r = -0.002$) and Spearman's rho is weakly negative ($\rho = -0.265$, $p = 0.012$). In contrast, donor programme volume is negatively associated with efficiency, with a Spearman correlation of $\rho = -0.491$. The more reasonable interpretation is that larger systems recover a broader and more medically heterogeneous donor pool, which increases overall activity but can reduce the share of donors whose organs are fully utilised. Accordingly, efficiency should not be interpreted as a stand-alone indicator of system quality. In methodological terms, this is another reason to distinguish between linear and rank-based association when interpreting the results.^[10,5]

5. Conclusions

The analysis yields three key conclusions. First, human development is closely tied to the capacity to generate deceased donors and perform transplants. Second, higher development is associated not just with more transplant activity, but also with a structural transition from living-donor and kidney-dominant systems toward deceased-donor and multi-organ systems. Third, HDI alone cannot explain national performance. Countries with similar development levels can still differ considerably because of policy design, institutional coordination, and cultural context. Finally, utilisation efficiency follows a different logic from donation capacity and should be interpreted in relation to programme scale rather than as a direct measure of quality.

References

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